How to work

- Keep the course files in the `stata/do/` folder. In Stata, move to it with `cd "path/to/stata/do"`: the folder must contain `data/`, `exercises/` and `00_setup.do`.
- Open the worksheet in `do/exercises/`, save it under a new name, and complete the TODO lines. Write your interpretations as comments (lines starting with `*`).
- Work in pairs, swapping who types and who explains; keep your own do-file and your own answers.
- Every number you write must come with its unit and the number of observations behind it. Simulated data are artificial and must never be described as findings.
- Laboratory work is formative and not graded.

Lab 1: contract enforcement across economies

Question. How long and how costly is it to enforce a commercial contract, and how does this vary across economies?

Data. `data/contracts_2019.dta`: World Bank, *Doing Business 2020*, enforcing-contracts indicators, data as of 1 May 2019 (source year label 2019), 191 economies, CC BY 4.0. One row is one economy. The indicators describe a standardized hypothetical commercial dispute in the largest business city, as estimated by local experts; they are not averages of observed court cases. Doing Business was discontinued in 2021 after data irregularities: treat these as historical teaching data.

Worksheet. `do/exercises/01_data_and_description.do`.

Tasks

1. Inspect the data (`describe`, `codebook`). What is one row? What are the units of `time_days` and `cost_pct_claim`?
2. Tabulate `region` and `income_group`. Write one sentence with a share and its denominator. How many economies score 0 on `court_automation`, and why is that not missing information?
3. Summarize `time_days` and `cost_pct_claim` (`summarize, detail`). Report mean, median, SD and quartiles with units. Why is the mean above the median?
4. Draw a histogram of `time_days` and one of its logarithm. How many observations does each use?
5. Locate Italy (or another economy) in the distribution: what share of economies is faster? Which phase (filing, trial, enforcement) accounts for most of its time?
6. Compare regions with `tabstat ..., by(region) statistics(n mean p50)`. Which region has the highest median time, and on how many economies is it based?
7. Write five sentences: (1) source, edition and unit; (2) one share with its denominator; (3) centre and spread of time; (4) one comparison; (5) one limitation.

Extension. `scatter trial_days judicial_quality`: describe the pattern in words only (no causal language).

Self-check

- Did I state that one row is an economy, not a case?
- Did every percentage come with its denominator?
- Did I avoid saying that "cases last 1,120 days on average" in Italy?

Lab 2: sampling, standard errors and confidence intervals

Question. If we had observed a different set of cases, how different would our estimate be?

Data. Entirely artificial. A supplied program draws samples of case durations from a right-skewed population with mean 300 days and SD 150 days, 2,000 times for each sample size $n = 5$, 25 and 100. For each sample it records the mean, the SD, the standard error $s/\sqrt{n}$ and a 95% confidence interval.

Worksheet. `do/exercises/02_sampling_and_inference.do` (it runs `do/ready_to_run/02_sampling_demo.do`).

Before class: reading

Gneezy, U. and Rustichini, A. (2000), A fine is a price, *Journal of Legal Studies* 29(1): 1–17. Free PDF. Read the abstract, Sections I–II, Table 1 (p. 6) and Figure 1 (p. 7); skip the formal tests and the model (30–45 minutes). Bring notes on:

1. What does one row of Table 1 record, and what does one cell count? What does Figure 1 average?
2. Which centres receive the fine, how were they chosen, and in which weeks is it in force?
3. Why are the 200 centre-week counts not 200 independent observations of the effect of the fine?

Tasks

1. Run the supplied simulation. What does one row of the results file represent?
2. For each n, compare: the average within-sample SD; the SD of the 2,000 sample means; the average estimated SE. Which falls as n grows, and by how much from 25 to 100?
3. Look at the three histograms of sample means. Where is the skew of the population still visible?
4. Compute the share of intervals that contain 300 for each n. Why is it not exactly 0.95? For which n is it furthest from 0.95, and why?
5. Extension: predict the SD of the sample means for $n = 400$, then check it with the commands in task 6 of the worksheet.
6. Write two paragraphs: (a) SD versus SE, with your numbers and units; (b) a correct interpretation of "95% confidence".
7. Paper bridge: three sentences on the unit of Table 1, the timing of the fine, and why repeated centre-weeks are not independent observations.

Self-check

- Did I avoid writing that 95% of cases fall inside the interval?
- Did I avoid writing that the true mean has a 95% probability of lying in my interval?
- Did I state clearly that the data are simulated?